INCLUSIONS OF THE ZARISKI, ETALE AND SMOOTH COVERING CLASSES IN THE SYNTOMIC COVERING CLASS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [1876, 1880] and proof(s) [[1882, 1887]]; this spec covers the complete environment instead (statement 1876-1880, proof span(s) [(1882, 1887)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. Definitional inclusion chain with cited facts.

Source reference. Original reference 01VD: sources/stacks-chow/tags.txt, source lines 2442–2442. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0220: sources/stacks-chow/tags.txt, source lines 2689–2689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Any smooth covering is a syntomic covering, and a fortiori, any étale or Zariski covering is a syntomic covering.*

Proof. This is clear from the definitions and the fact that a smooth morphism is syntomic, see Morphisms, Lemma 01VD and Lemma 0220. □

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